

Predicting Gold Reserve Diversification in BRICS and Emerging Markets

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a. Introduction

Central banks hold international reserves to support external stability, manage exchange-rate pressures, and protect against financial shocks. Although foreign currency assets make up the majority of most reserve portfolios, gold remains an important reserve asset because it is not directly tied to the liabilities of another sovereign issuer and may serve as a hedge against macroeconomic, geopolitical, and financial uncertainty. This project examines whether gold reserve shares in BRICS and selected emerging-market economies can be predicted using lagged macroeconomic and institutional variables. Using World Bank World Development Indicators data from 1996 to 2022, I construct a country-year panel measuring the share of total reserves held in gold and compare OLS, ridge, and lasso regression models across three specifications: reserve-share levels with lagged gold share, reserve-share levels without lagged gold share, and year-to-year changes in gold share. The central question is whether reserve composition is primarily explained by persistence or whether lagged economic and institutional conditions help predict future diversification behavior.

b. Data Description

As mentioned prior, my data was compiled from the World Development Indicators (WDI). The working sample includes the original BRICS countries and selected emerging markets, with data from years 1996-2022. The specific countries are Brazil, Russia, India, China, and South Africa, along with Mexico, Argentina, Chile, Colombia, Peru, Turkey, Poland, Hungary, Romania, the Czech Republic, Ukraine, Kazakhstan, Saudi Arabia, the United Arab Emirates, Qatar, Egypt, Morocco, Nigeria, Kenya, Ghana, Ethiopia, Indonesia, Malaysia, Thailand, Vietnam, the Philippines, Pakistan, Bangladesh, and Sri Lanka. These were chosen to provide a regionally representative sample of emerging markets globally.

After constructing lags and dropping rows with missing values as needed for modeling, the effective sample used in the main specifications contains 475 training observations (1997–2018) and 116 test observations (2019–2022).

My outcome variable is `gold_share`, representing the percent of total reserves held as gold. This is constructed as $100 \times (\text{gold_reserves} / \text{total_reserves})$, where $\text{gold_reserves} = \text{total_reserves} - \text{reserves_excluding_gold}$.

My predictors (lagged by 1 year to ensure the changes outcome are driven by prior time period effects) are the prior year outcome to control for persistence effects (gold_share_lag1), GDP per capita (gdp_pc_lag1), GDP per capita growth, annual % (gdp_g_lag1), inflation, consumer prices, annual % (infl_lag1), trade as % of GDP (trade_lag1), manufacturing value added as % of GDP (manuf_lag1), fuel exports as % of merchandise exports (fuel_lag1), domestic credit to private sector as % of GDP (credit_lag1), political stability and absence of violence/terrorism estimate (polstab_lag1), and a BRICS vs other EM's dummy (brics_dummy). The predictors include both a variety of continuous variables (inflation, GDP per capita, etc.) and one dummy variable.

The WDI export provided a file format with year columns in wide format and values stored as text. As a result, I had to include cleaning processes to detect for year columns, reshape wide-to-long and then long-to-wide to build a panel, convert indicators to numeric, filter to ensure plausible gold_share values (0–100), and construct one-year lags within country.

I engineered 4 features: (1) gold_share (constructed from two WDI reserve series), (2) brics_dummy (constructed from country codes), (3) one-year lags of all predictors and the outcome, and (4) the change in gold share for Spec 3.

Before turning to the predictive models, I begin with some background on gold reserve shares across countries. Figure 1 shows the distribution of gold shares for BRICS versus non-BRICS emerging markets over the full time period. This allows for a visual assessment of differences and similarities in tendency and dispersion between the two groups and motivates the inclusion of a BRICS dummy in the forecasting models. To complement the cross-sectional comparison, *Figure 2* plots the average gold share over time for BRICS and non-BRICS countries. This figure highlights how reserve composition has evolved from 1996 to 2022 and whether the groups exhibit different trends or structural shifts. The figure illustrates that there have been notable changes in reserve composition throughout the period for both groups. Examining these patterns provides context for understanding persistence and post-2008 strategic adjustments.

Figure 1

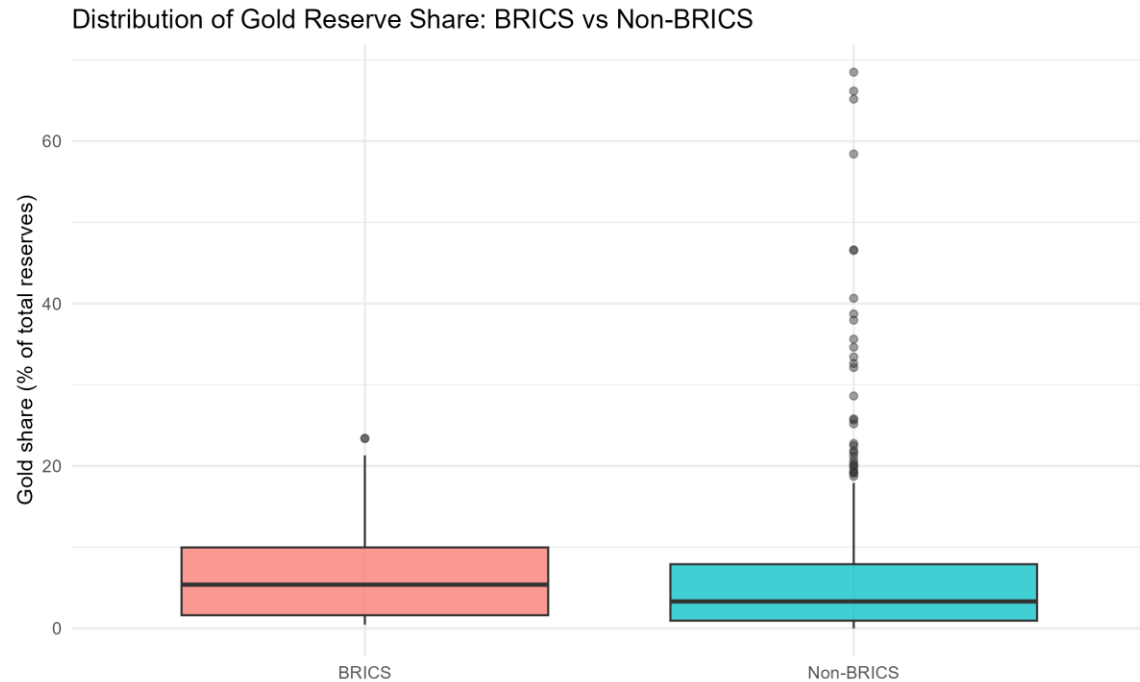


Figure 1 shows the distribution of gold reserve shares for BRICS and non-BRICS emerging economies. BRICS countries exhibit a higher median gold share and a wider dispersion compared to non-BRICS countries. This suggests that BRICS members, particularly Russia and China, may rely more heavily on gold as part of their reserve composition strategy. The greater variability among BRICS also indicates heterogeneity in reserve management within the group.

Figure 2



Figure 2 plots the average gold reserve share for BRICS and non-BRICS countries over the sampled years. Trends for both are similar across time, with both groups displaying a negative trend in gold share from 1996-2007 and a reversal to a positive trend in gold share since 2008. This change in trend likely reflects strategic reserve diversification following the global financial crisis. Overall, the time trend suggests that BRICS countries have increasingly differentiated themselves in reserve composition.

c. Methods

To mimic forecasting, I use a time-based split. The training set includes years 1996-2018 and the test set includes years 2019–2022. All model fitting and hyperparameter tuning are performed using the training data only.

I report root mean squared error (RMSE) and mean absolute error (MAE) on the held-out test set. RMSE penalizes large errors more heavily; MAE is more robust to outliers and is slightly easier to interpret as the typical absolute error in percentage points of `gold_share`.

I compare the following model types:

1) OLS regression: The standard linear regression estimated by least squares. It provides an interpretable baseline but can overfit when predictors are correlated or when the signal in the model is weak relative to the noise.

2) Ridge regression: A regularized linear model that shrinks coefficients toward zero using an L2 penalty. Ridge is useful when predictors are correlated with each other because it stabilizes estimates, improving out-of-sample performance.

3) Lasso regression: A regularized linear model that uses an L1 penalty. Lasso can set some coefficients exactly to zero, providing forced feature selection and potentially improving interpretability and prediction when only a small number of predictors matters.

For ridge and lasso, I use `cv.glmnet` on the training set (`glmnet` uses 10-fold CV by default) to select the penalty parameter λ . I use `lambda.min` to find the λ that minimizes mean CV error and print the corresponding CV RMSE.

Specifications:

Spec 1: For my first specification, I predict `gold_share` using `gold_share_lag1` plus lagged predictors and `brics_dummy`. This specification controls for what I presume will be strong persistence in reserve composition.

Spec 2: For my second, I predict `gold_share` using only the lagged predictors and `brics_dummy`. This tests whether macro/institutional variables alone can explain levels of `gold_share`.

Spec 3: Lastly, I predict $\Delta \text{gold_share} = \text{gold_share} - \text{gold_share_lag1}$ using lagged predictors and `brics_dummy`. The purpose of this third version is to test whether macroeconomic variables are able to better explain shifts as compared to levels.

Baselines: I report (1) a persistence test that predicts `gold_share` simply with `gold_share_lag1` and (2) another baseline that predicts the training-set mean `gold_share` for all test observations, ignoring all country and time based variation.

d. Results

Table 1 summarizes performance on the held-out test set (2019–2022) and, for ridge/lasso, the cross-validated RMSE within the training set.

	Train RMSE OLS	Train RMSE Ridge	Train RMSE Lasso	Test RMSE OLS	Test RMSE Ridge	Test RMSE Lasso	CV RMSE Ridge	CV RMSE Lasso
Spec 1: Level + Lagged Y	2.818	2.859	2.844	5.162	5.777	5.363	3.067	3.026

Spec 2: Level no Lagged Y	5.853	5.881	5.861	13.677	13.665	13.673	6.028	5.957
Spec 3: Change in Gold Share	3.190	3.209	3.209	4.077	4.099	4.099	3.213	3.225

Baseline performance, evaluated on the test set:

- Lag-only baseline: Test RMSE = 4.060, Test MAE = 1.728.
- Train-mean baseline: Test RMSE = 13.979, Test MAE = 7.435.

I include both persistence and mean baselines to see whether the predictive models outperform simple forecasting and to quantify the contribution of my macroeconomic predictors beyond basic inertia.

Key conclusions:

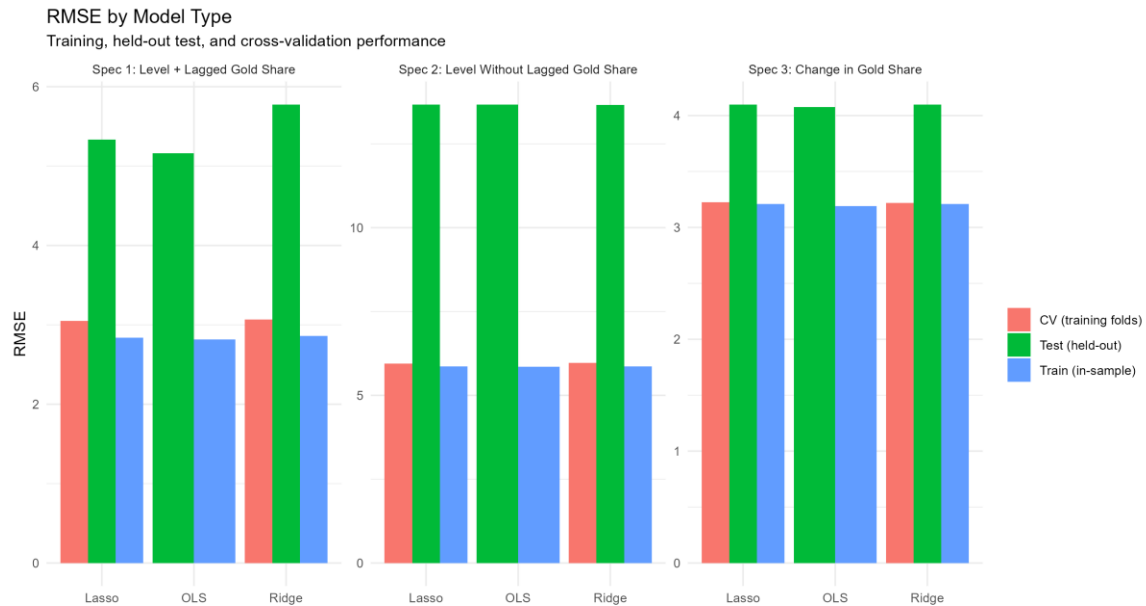
1) Spec 1 (levels + lagged outcome) fits the training data well but does not beat the simple lag-only baseline on the test data. This suggests that most of the predictive power comes from persistence rather than the additional predictors included in the model.

2) Spec 2 (levels without lagged outcome) performs poorly out of sample, only marginally better than the mean baseline and significantly worse than the lag-only baseline. This indicates that macro/institutional predictors do not predict the level of gold_share well by themselves.

3) Spec 3 (changes) performs best in terms of test RMSE. Because Spec 3 predicts changes rather than levels, its RMSE is not directly comparable in magnitude to level specifications, but it indicates improved generalization. The typical prediction error is about 4 percentage points in the year-to-year change of gold_share. The similarity in outcomes between OLS, ridge, and lasso suggests that regularization is not adding much predictive power to the model as is.

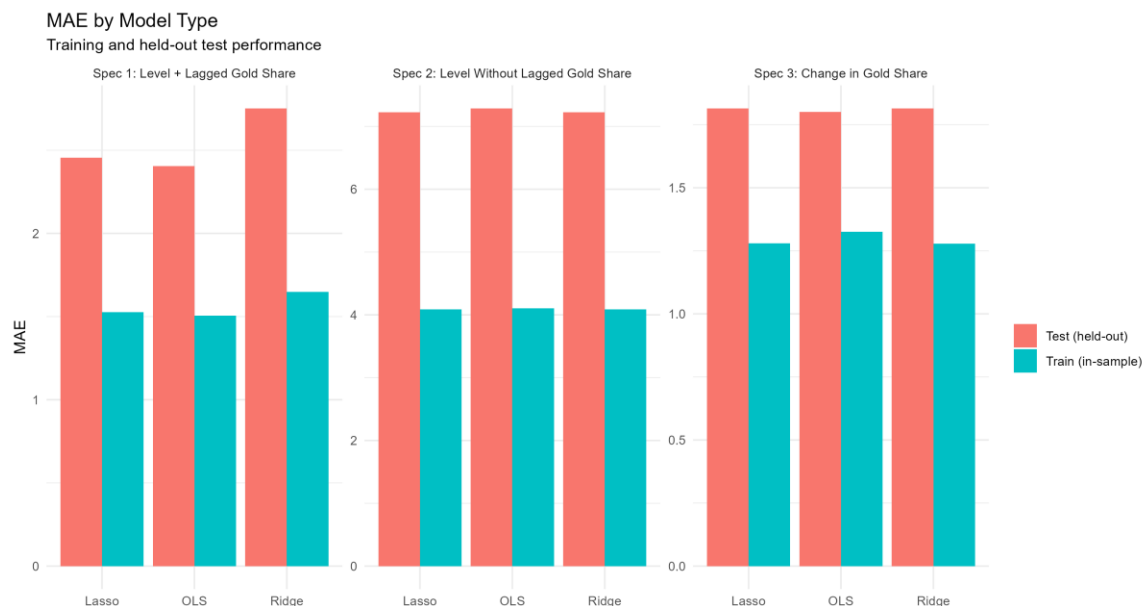
4) The effects of regularization are minimal. While ridge and lasso do have slightly higher test RMSE than OLS in Specs 1 and 3, the differences are small even in those cases. This may be because the model already has relatively few predictors, and the main issue is weak predictive signal rather than overfitting.

Figure 3



The RMSE comparison shows that models including lagged gold share outperform those excluding it, confirming strong persistence in reserve composition. Spec 3 achieves the lowest test RMSE, showing improved generalization. Ridge and Lasso exhibit similar performance to OLS, suggesting that overfitting is not the primary challenge in this relatively simple model. Cross-validation errors align closely with test errors, providing evidence that the training procedure was stable and effective.

Figure 4



The MAE results reinforce the RMSE findings, with Spec 3 again producing the smallest out-of-sample prediction error. The similarity between RMSE and MAE rankings suggests that results are not driven by extreme outliers. Models excluding lagged gold share perform worse, showing the importance of persistence. Overall, penalized methods provide only modest improvements, indicating that the predictive signal is limited rather than overly complex.

e. Conclusion

This project examined whether gold reserve shares in BRICS and selected emerging-market economies can be predicted using lagged macroeconomic and institutional variables. Using World Bank WDI data from 1996 to 2022, I constructed a country-year panel and compared OLS, ridge, and lasso models across level and change specifications. The results show that gold reserve composition is highly persistent. For level forecasts, the strongest benchmark was simply the prior year's gold share, and adding lagged macroeconomic variables and a BRICS indicator did not meaningfully improve out-of-sample performance relative to that persistence baseline.

The change specification provides a more useful interpretation of the macroeconomic predictors. Although year-to-year changes in gold reserve share remain difficult to forecast precisely, the results suggest that lagged economic and institutional variables may be more informative for explaining adjustments in reserve composition than for predicting long-run reserve-share levels. The similarity between OLS, ridge, and lasso performance also suggests that the main limitation is not overfitting in a high-dimensional model, but rather the limited predictive signal available in the selected annual macroeconomic variables.

Future work could improve the analysis by incorporating additional variables that more directly capture reserve-management incentives, including exchange-rate movements, commodity prices, sanctions exposure, geopolitical risk measures, central-bank policy indicators, and IMF reserve-composition data. Longer lag structures or rolling averages may also be appropriate because reserve allocation decisions often unfold gradually rather than within a single year. Overall, the project finds that persistence dominates reserve-share levels, while macroeconomic and institutional conditions may be more useful for understanding changes in reserve-diversification behavior over time.